"Ch:1" "Chem 101"

Classification of matter

(*) we define matter as anything that has (mass) and takes up space.

(*) المادة :- اي شيء له كتلة و يشغل حيز

(*) Atoms :- are the building blocks of matter.

الذرات هي المكون الاساسي للمادة

(*) element :- is made of the same kind of atom.

العناصر تتكون من نفس النوع من الذرات

(*) Compound :- is made of two or more different kinds of elements.

المركب يتكون من عنصرين او اكثر

(*) state of matter :- حالات المادة

1) Solid :- صلبة

خصائص للحالة الصلبة :-

(*) ordered arrangement, Particles are essentially in fixed Positions, Particles close together

2) Liquid :- سائلة

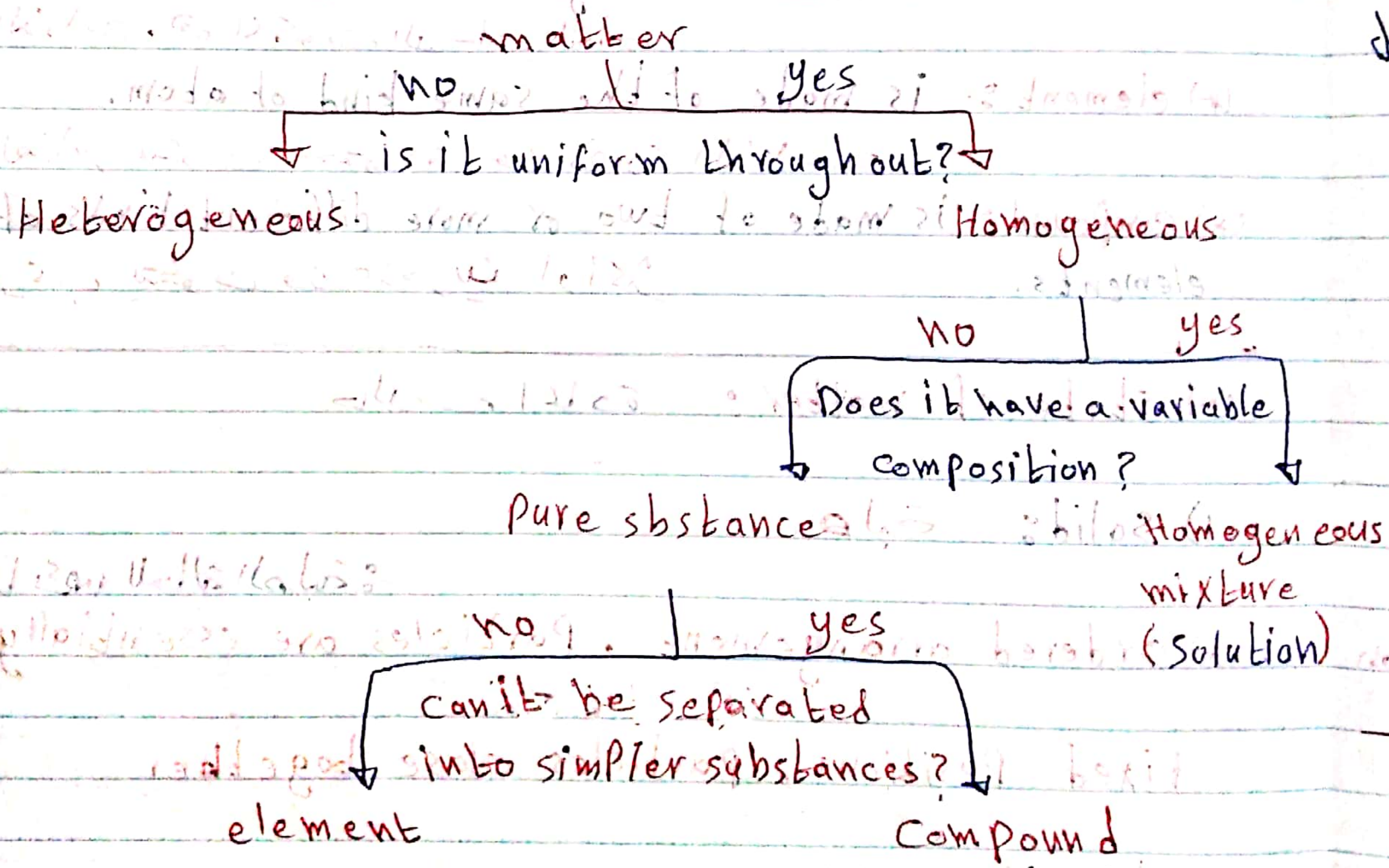
خصائص للحالة السائلة :-

(*) Disorder, Particles are free to move relative to each other, Particles close together.

3) Gas :- غازية

(*) total disorder, much empty space, particles have freedom of motion, particles far apart.

$$Gas \underset{heat}{\overset{cool}{\rightleftarrows}} Liquid \underset{heat}{\overset{cool}{\rightleftarrows}} solid$$



هاي هي بشكل سريع تصنيفات المادة المختلفة

Properties of matter:
Types of properties

1) Physical properties :- خصائص فيزيائية

- can be observed without changing a substance into another subsance.

مثل :- Boiling point , density , mass , volume , etc ---

2) chemical properties :- خصائص كيميائية

- can only be observed when a substance is changed into another substance.

مثل :- flammability , corrosiveness , reactivity with acid , etc ---

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هناك ← خاصيات اخرى خلونا نشوفهم :-

(*) intensive properties :-

- are independent of the amount of the substance

خاصيه لا تعتمد على كمية المادة يعني لو كنتك واحد لتر من الماء

و كثرة لتر من الماء تبقى درجة غليانهم عند 100° سيليسوس

مثل :- Density, boiling point, color, etc —

(*) extensive properties :-

- Depend upon the amount of the substance.

خاصية تعتمد على كمية المادة

مثل :- mass, volume, energy, etc —

(*) types of changes :- انواع التغيرات

1) Physical changes :- don't change the composition

مثل :- changes of state, temperature, volume, etc —

2) chemical changes :- result new substance

مثل :- Combustion, oxidation, decomposition, etc —

Unites of measurement

((SI Units))

1) mass :- "kg"
2) Length :- "m"
3) Time :- "S"
4) Temperature :- "K"
5) Amount of substance "mol"
6) electric current "A"
7) Luminous intensity "cd"

"prefixes"

1) Peta P 10^{15}
2) Tera T 10^{12}
3) Giga G 10^{9}
4) Mega M 10^{6}
5) kilo k 10^{3}
6) Deci d 10^{-1}
7) Centi c 10^{-2}
8) Milli m 10^{-3}
9) Micro μ 10^{-6}
10) Nano n 10^{-9}
11) Pico p 10^{-12}
12) Femto f 10^{-15}
13) Atto a 10^{-18}
14) Zepto z 10^{-21}

<u>example</u>:- How many femto-meters are present in 1m?

$$1\ fm \rightarrow 10^{-15}\ m$$

$$So\ 1\ m \rightarrow 10^{15}\ fm$$

<u>example</u>:- How many nanograms are there in a picogram?

$\boxed{\text{الناتج}}$

$\boxed{\text{الأس المعطى}} - \boxed{\text{الأس المطلوب}} = \text{الرقم} \times 10$

$-12 - (-9) = -3$

الأس تبع البيكو المعطى

1×10^{-3}

الأس تبع النانو المطلوب

$So\ 1 pg = 10^{-3}\ ng$

Temperature

كيلفن — سيليسيوس — فهرنهايت

(*) $K = C^\circ + 273$

(*) $F^\circ = \frac{9}{5} C^\circ + 32$ حفظ مهم

(*) $C^\circ = \frac{5}{9} (F^\circ - 32)$

example:- Convert $28^\circ C$ to the kelvin scale

$$K = C^\circ + 273$$

$$K = 28 + 273 = 301\ K$$

example:- Convert $28^\circ C$ into the corresponding fahrenheit.

$$F^\circ = C^\circ * \frac{9}{5} + 32$$

$$F^\circ = 28 * \frac{9}{5} + 32 = 82\ F^\circ$$

Volume:

(*) $1 mL = 1 cm^3$

(*) $1 L = 1 dm^3$

(*) $1 m^3 = 1000 L = 1000 dm^3 = 10^6 cm^3$

(*) $1 L = 1000 cm^3$

Density

Density ← $d = \frac{mass (g)}{Volume (mL \text{ or } cm^3)} = \frac{g}{cm^3}$

example:- The density of piece of copper wire is $8.96 g/cm^3$. Calculate the volume in cm^3 of a piece of copper with a mass of $4.28 g$

$$d = \frac{m}{V}$$

$$8.96 = \frac{4.28}{V}$$

Volume ← $V = \frac{4.28}{8.96} = 0.478 cm^3$

accuracy and Precision

(*) accuracy :- Refers how to closely individual measurement agree with true value.

(*) precision :- is a measure of how closely individual measurement agree with one another.

Significant figures

الأرقام المعنوية

1) All nonzero digits are sig.fig

423.444 → 6 sig.fig

32156 → 5 sig.fig

2) zeroes between nonzero digits are sig.fig

42300045 → 8 sig.fig

42340.0025 → 9 sig.fig

3) zeroes at the end of a number are sig.fig if a decimal point is written in the number.

423000.000 → 9 sig.fig

423 000 ⟶ 3 Sig. fig

4) Zeroes at the beginning of a number are <u>never</u> sig. fig

00042345 ⟶ 5 sig. fig

0.00048 ⟶ 2 Sig. fig

5) numbers ends with zeroes but contain no decimal point, then zeroes are <u>not</u> sig. fig

400 ⟶ 1 sig. fig

10300 ⟶ 3 Sig. fig

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example :- Determine the number of sig. fig in each of the following :-

1) 345.5 ⟶ 4 sig. fig

2) 0.0058 ⟶ 2 Sig. fig

3) 1204 ⟶ 4 Sig. fig

4) 250 ⟶ 2 Sig. fig

5) 250.00 ⟶ 5 Sig. fig

Addition and subtraction

: when addition or subtraction performed ; answers are rounded to the least significant decimal place

example :- $102.50 + 0.231$

2 decimal point — 3 decimal point

الجواب لازم يكون فيه اقل decimal point

$= 102.731 \rightarrow$ rounding $\rightarrow 102.73$

example :- $12.11 + 18.0$

2 decimal point — 1 decimal point

الجواب لازم يكون فيه اقل decimal point

$= 30.11 \rightarrow$ rounding $\rightarrow 30.1$

Multiplication and division

- final answer contain the smallest number of significant figures.

example :-

1.4 × 8.011 = 11.2154 → rounding to 11

1.4 → 2 sig.fig
8.011 → 4 sig.fig
11 → 2 sig.fig

example :- 4.567 × 1.4 × 3.00 = 19.1814

4.567 → 4 sig
1.4 → 2 sig
3.00 → 3 sig

لازم يكون الناتج اقل sig.fig

19.1814 → rounding → 19

كيف تعمل rounding :-

(*) الرقم المراد التخلص منه اكبر او يساوي 5 نضيف 1 على اليسار

نضيف واحد ← 11.712 → 12

(*) الرقم المراد التخلص منه اقل من 5 لا نضيف اشي على اليسار

لا نضيف واحد ← 1.23 → 1